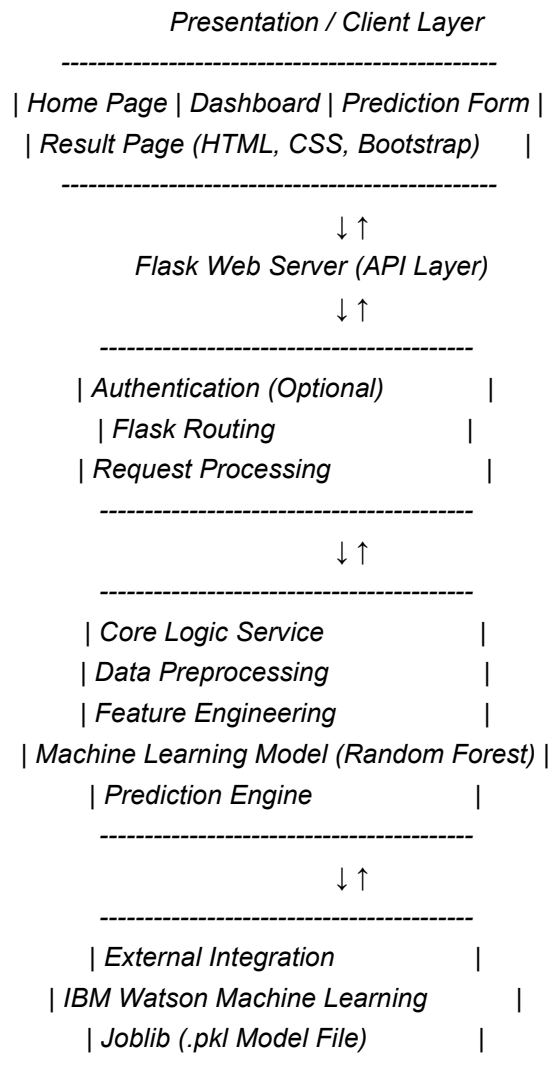


Solution Architecture

Date	4 july,2026
Team ID	SWTID-2026-5594
Project Name	CREDIT CARD APPROVAL PREDICTION SYSTEM
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface where applicants enter their personal and financial details and view prediction results.	HTML5, CSS3, Bootstrap 5, JavaScript
API Layer (Flask Server)	Handles user requests, validates input, communicates with the machine learning model, and returns prediction results.	Python, Flask
Core Logic Service	Performs data preprocessing, feature encoding, loads the trained Random Forest model, and generates approval/rejection predictions.	Scikit-learn, Random Forest, Pandas, NumPy, Joblib
External Integration	Supports cloud deployment and model integration.	IBM Watson Machine Learning
Database / Data Storage	Stores the datasets used for training and the trained machine learning model used during prediction.	CSV Files, Joblib (.pkl)